3 **(a)** A balloon of mass 15 g is glued to a straw. The straw is threaded onto a horizontal string, as shown in Fig. 3.1.
The balloon is filled with air and then the air is released.

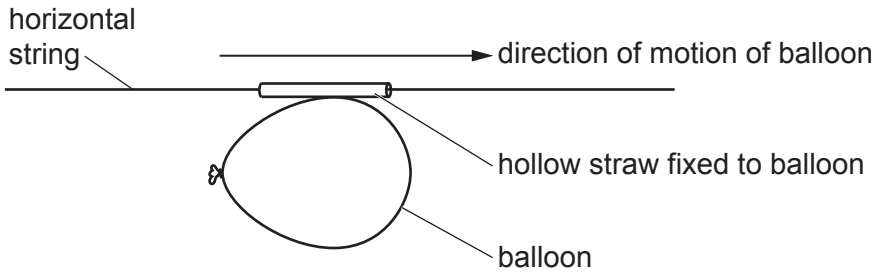


Fig. 3.1

As the air leaves the balloon, the balloon experiences a force.
The balloon accelerates from rest until it reaches a constant speed. It then travels 0.67 m in 0.18 s at this constant speed.

(i) Explain in words what is meant by the term impulse.

.....

..... [1]

(ii) Calculate the resultant impulse on the balloon while it is accelerating.

impulse = ... [3]

(iii) Explain how momentum is conserved as the balloon accelerates.

.....

.....

..... [2]

(b) Fig. 3.2 shows the directions of two forces acting on a different balloon as it moves.

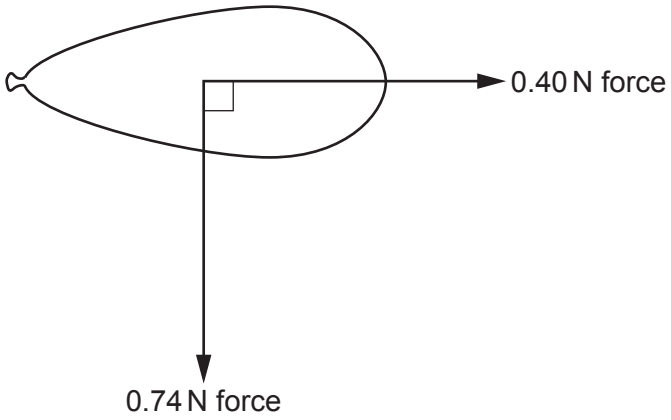


Fig. 3.2 (not to scale)

Determine the magnitude and direction of the resultant force on the balloon.

magnitude .....

direction relative to horizontal force .....
[4]

[Total: 10]